

# Takeoff Jet Noise analysis for a Turbofan Engine



Advisor : Dr. Roy Myose Ph.D.

Garrett TFE731 Engine on display at National Air and Space Museum in Washington, DC [5]

# OBJECTIVES

1. Complete a literature review on Airplane Acoustics and Turbofan Performance such that
  - a) Formulate an analytical method to calculate/estimate the takeoff sound intensity.
  - b) Find a turbofan engine noise characteristics report.
    - a) Document the engine specification and parameters needed to perform the analytical analysis.
    - b) Document the sound intensity during takeoff from the report.
2. Determine the total sound intensity level (in db.) using analytical methods.
  1. Calculate core and fan exhaust velocity.
  2. Calculate the sound intensity based on the exhaust velocities and exhaust areas.
3. Compare the Results obtained from Analytical method with the test data.
  1. Provide a comparison chart for the sound Intensity characteristics.
  2. Provide a discussion/explanation on the result.

# Literature Review

# Analytical Method

- The Analytical method used in this project will be leveraged from Dr. Myose's AE742 lecture. Hill & Peterson [1] provides the fundamentals of thermodynamics that frames the basis for this analytical method.
- The final relationships from the method are outlined in the next two slides.
- A detailed class project previously done by the author used this method to do a Turbofan Sizing of a futuristic turbofan jet and it's acoustic Analysis. Appendix 1.i

# Method – Using Inlet Area to calculate T04

- The fan diameter and the maximum Thrust was leveraged to calculate the maximum cycle temperature.
  - $a_{total} = (1 + \beta) * a_{02}$  ( $a_{02}$  = coreinlet area)
    - $a_{total} = \pi D^2$  ( $D$  is the fan diameter)
- Vary the T04 till the required core inlet area ( $a_{02}$ ) is reached.

## Method - Calculate exhaust velocity[9]

$$u_{\infty} = M_a \sqrt{\gamma_D R T_a}$$

$$T_{02} = T_a \left[ 1 + \frac{1}{2} (\gamma_D - 1) M_a^2 \right]$$

$$P_{02} = P_a \{ 1 + \eta_D [(T_{02}/T_a) - 1] \}^{\frac{\gamma_D}{\gamma_D - 1}}$$

$$\rho_{02} = P_{02} / (R T_{02})$$

$$U_{02} = M_{02} \sqrt{\gamma_D R T_{02}} = 0.3 \sqrt{\gamma_D R T_{02}}$$

$$T_{03} = T_{02} \left[ 1 + \frac{1}{\eta_C} (P_{rc} (\gamma_C - 1) / \gamma_C - 1) \right]$$

$$f = \frac{T_{04} - T_{03}}{(Q_R / c_{p,B}) - T_{04}}$$

$$T_{08} = T_{02} \left[ 1 + \frac{1}{\eta_F} (P_{rF}^{(\gamma_F - 1)/\gamma_F} - 1) \right]$$

$$T_{05} = T_{04} - \left( \frac{1}{1 + f} \right) \left( \frac{h_{comp}}{c_{p,T}} \right) - \left( \frac{\beta}{1 + f} \right) \left( \frac{h_{Fan}}{c_{p,T}} \right)$$

$$P_{05} = P_{04} \left[ 1 - \frac{1}{\eta_T} \left( 1 - \frac{T_{05}}{T_{04}} \right) \right]^{\frac{\gamma_T}{\gamma_T - 1}}$$

$$u_{e,C} = \sqrt{2 \eta_N \frac{\gamma_N R}{\gamma_N - 1} T_{05} \left[ 1 - \left( \frac{P_a}{P_{05}} \right)^{\frac{\gamma_N - 1}{\gamma_N}} \right]}$$

$$u_{e,F} = \sqrt{2 \eta_{N,F} \frac{\gamma_{N,F} R}{\gamma_{N,F} - 1} T_{08} \left[ 1 - \left( \frac{P_a}{P_{08}} \right)^{\frac{\gamma_{N,F} - 1}{\gamma_{N,F}}} \right]}$$

$$T / \dot{m}_a = (1 + f) u_{e,C} + \beta u_{e,F} - (\beta + 1) u_{\infty}$$

$$\text{Core } \dot{m}_a = 16458 / \left( \frac{T}{\dot{m}_a} \right)$$

$$\text{Core area } A_{02} = \frac{\dot{m}_a}{(\rho_{02} U_{02})}$$

$$\text{TSFC} = f / (T / \dot{m}_a)$$

$$\left( 2 \eta_N \frac{\gamma_N}{\gamma_N - 1} T_{05} \left[ 1 - \left( \frac{P_a}{P_{05}} \right)^{\frac{\gamma_N - 1}{\gamma_N}} \right] \right)$$

# Method - Calculate the sound intensity based on the velocity and area[9]

- Sound Intensity Level (db) =  $10 \cdot \log_{10}(I/I_0)$
- where threshold of hearing  $I_0 = 10^{-12} \text{ W/m}^2 = 10\text{db}$

- Sound Intensity  $I = \rho / \left[ 4\pi \left( r_{\text{sphere}} \right)^2 \right]$

- Low Speed Acoustic Power for  $\frac{u_e}{a_0} < 2$ :

- $\rho = 10^{-4} \rho_0 u_e^8 A_e / a_0^5$  where  $A_e = \pi D^2 / 4$  &  $a_0 = \sqrt{\gamma R T_a}$

- High Supersonic Acoustic Power  $\frac{u_e}{a_0} > 2$

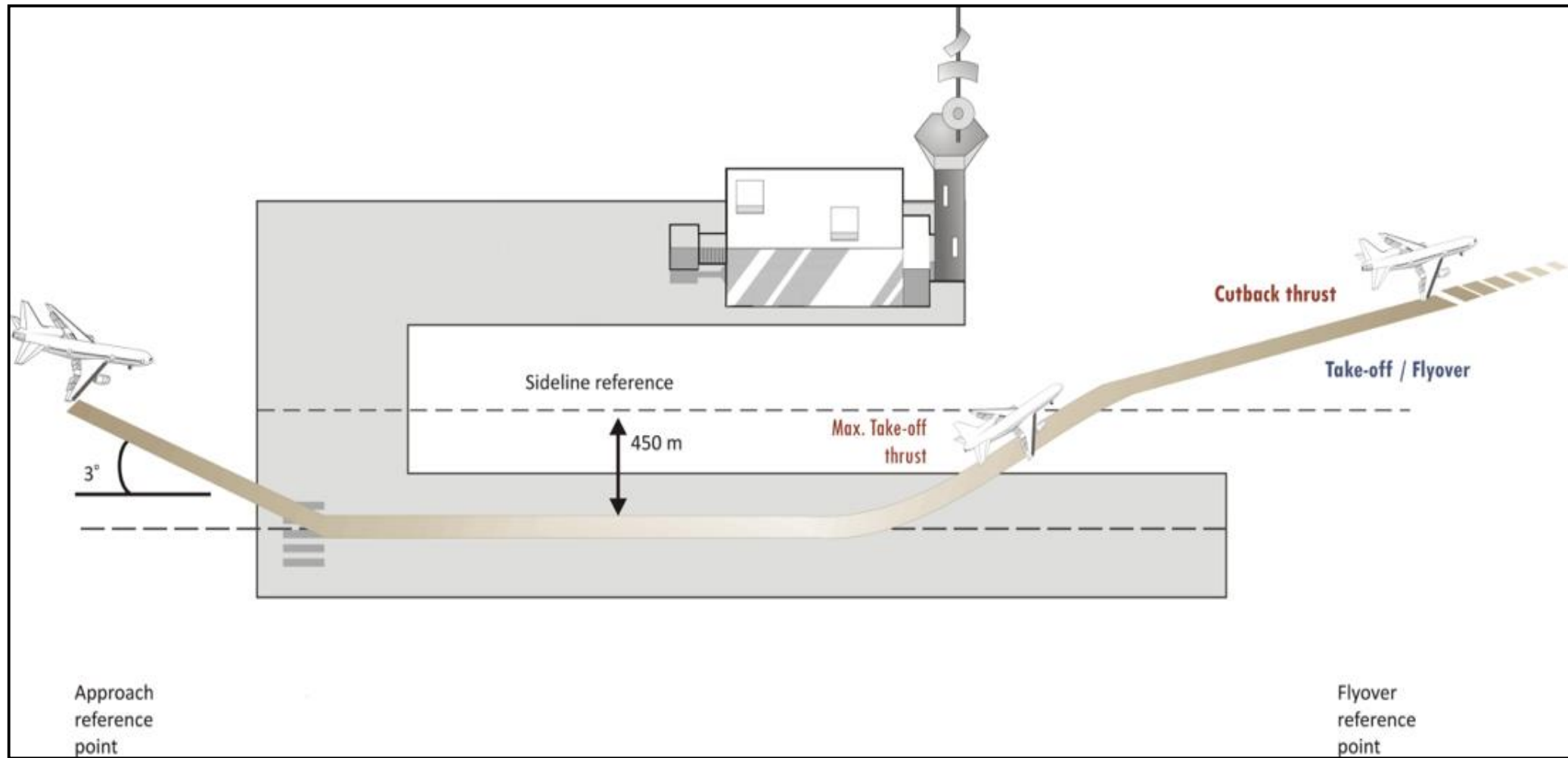
- $\rho = 0.003 \rho_0 M_e^3 A_e a_0^3$  where  $M_e = u_e / a_e$

- Multiple Noise Sources:  $I = 10 \log_{10} \left\{ \Sigma [10^{I_i / I_0}] \right\} = 10 \log_{10} \left\{ 10^{I_1 / I_0} + \dots + 10^{I_n / I_0} \right\}$

- $A_e$  for core =  $A_{02}$  ,  $A_e$  for fan =  $\beta * A_{02}$

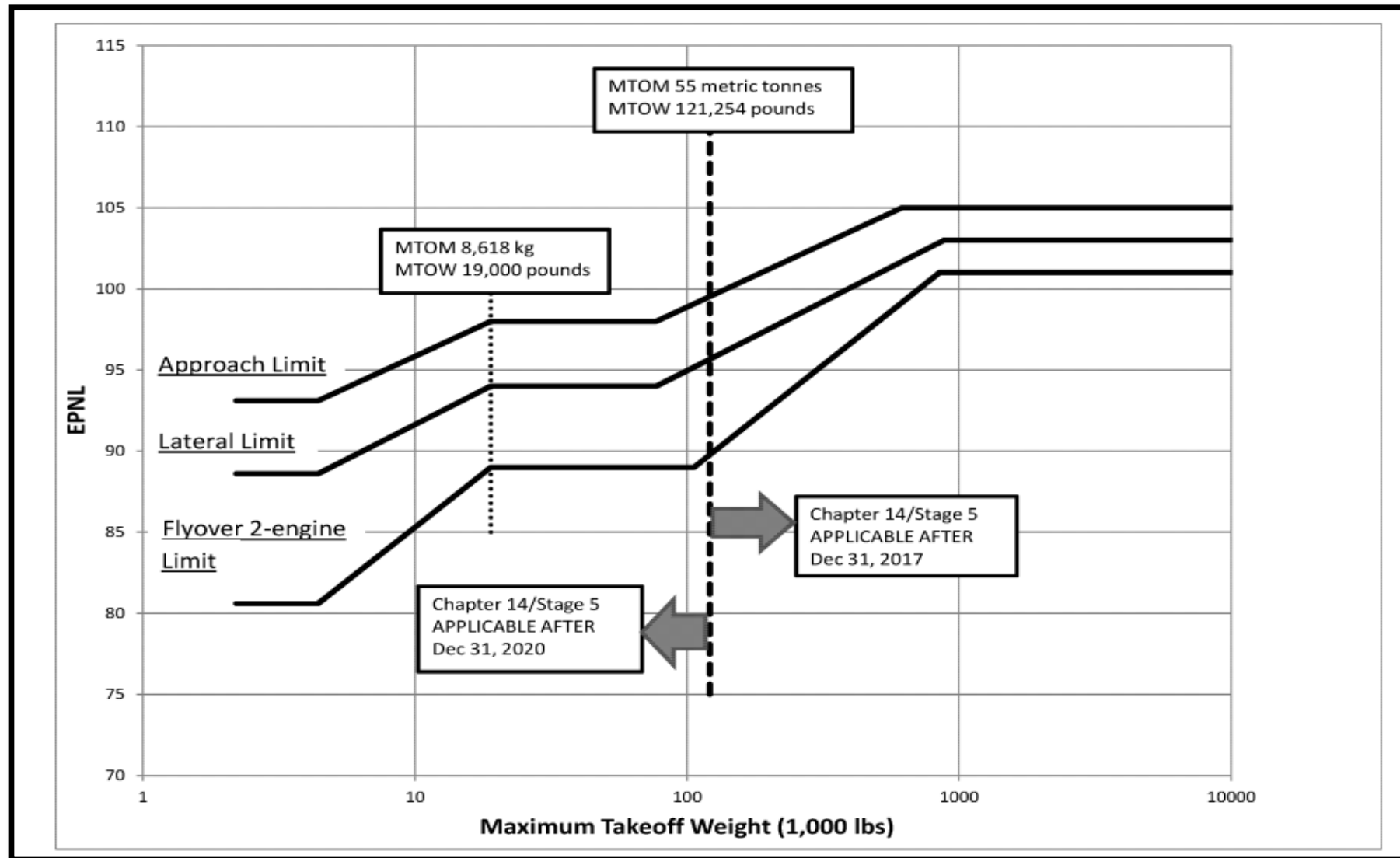
- $a_e$  is calculated by calculating  $T_7$  for the Core for high speed.  $T_7 = T_{05} \left\{ 1 - \eta_N \left[ 1 - \left( \frac{P_a}{P_{05}} \right)^{(\gamma_N - 1) / \gamma_N} \right] \right\}$

# Aircraft noise certification reference measurement points[7]

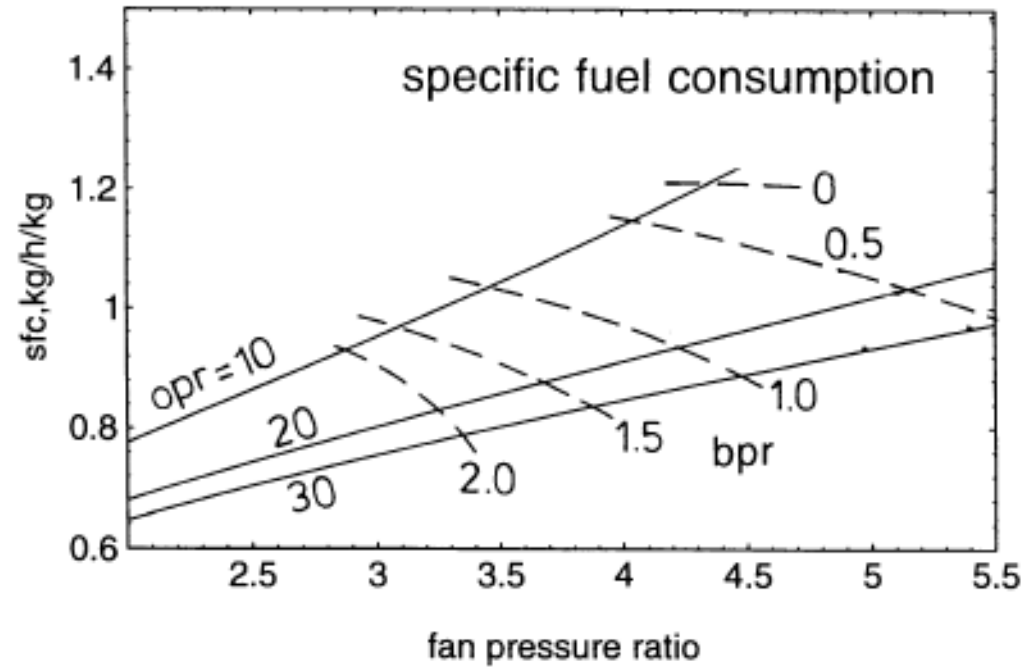




# Representation of Noise Limits as calculated at each microphone locations [8]



# Method - Fan Pressure Ratio Variation [3]



- Varied the Pressure ratio ( 1.5, 1.6, 1.7, 1.8 , 1.9 and 2.0)
- Compared the sound intensity with varied fan ratios outlined above.

# Garrett TFE731-3

## Turbofan

# Garrett TFE731 Engine Specification

| Parameters       | Values                |
|------------------|-----------------------|
| Thrust           | 3700 lbs.             |
| Fan Diameter     | 28.2 inches [4]       |
| Core             | Length : 49.75 inches |
| BPR              | 2.8                   |
| Compressor       | 1C stage LPC          |
|                  | 4 stage HPC           |
|                  | Pressure Ratio : 14.6 |
| Combustor        |                       |
| Turbine          | 1 stage HPT           |
|                  | 3 stage LPT           |
| Cruise Condition | Mach No : 0.8         |
|                  | Altitude : 40,000     |
| Weight (Dry)     | 775 lbs.              |

Table 3 : TFE731 Engine Specification

The Engineering report didn't contain any engine specifications that would be needed to do the flow calculations. The official Honeywell website provided the takeoff thrust[5], and the Smithsonian Website[6] provided the rest of the parameters that is provided on Table 3 above.

# Garrett TFE731-3A Noise characteristics results

| Flight Condition | EPNdB<br>TFE731-3A/Lear 50 |
|------------------|----------------------------|
| Takeoff          | 84.0                       |
| Sideline         | 86.9                       |
| Approach         | 92.4                       |

Table 1 : TFE731-3A/Learjet 50 series flyover noise[2]

| Noise Source  | EPNdB<br>TFE731-3A |
|---------------|--------------------|
| Fan Inlet     | 56.3               |
| Fan Discharge | 62.2               |
| Combustion    | 63.5               |
| Jet           | 82.5               |
| Turbine       | 49.2               |
| Total         | 84.0               |

Table 2 : Take off Noise comparison[2]

# Takeoff Acoustic Power Measurement Condition[2]

| Airspeed  | LPC Speed  | Temperature(Ambient) | Pressure(Ambient)             |
|-----------|------------|----------------------|-------------------------------|
| 168-knots | 18,261 rpm | 60F<br>288.706 K     | 28.54 inch hg<br>96647.533 Pa |

The take off condition is provided in the section 6.1 of the [2] which is summarized above , the acoustics labs Reports in the appendix ,[2] Appendix Lab Run #7 provides the Maximum Temperature of 1580F (1133.15 K) (marked as T5 )

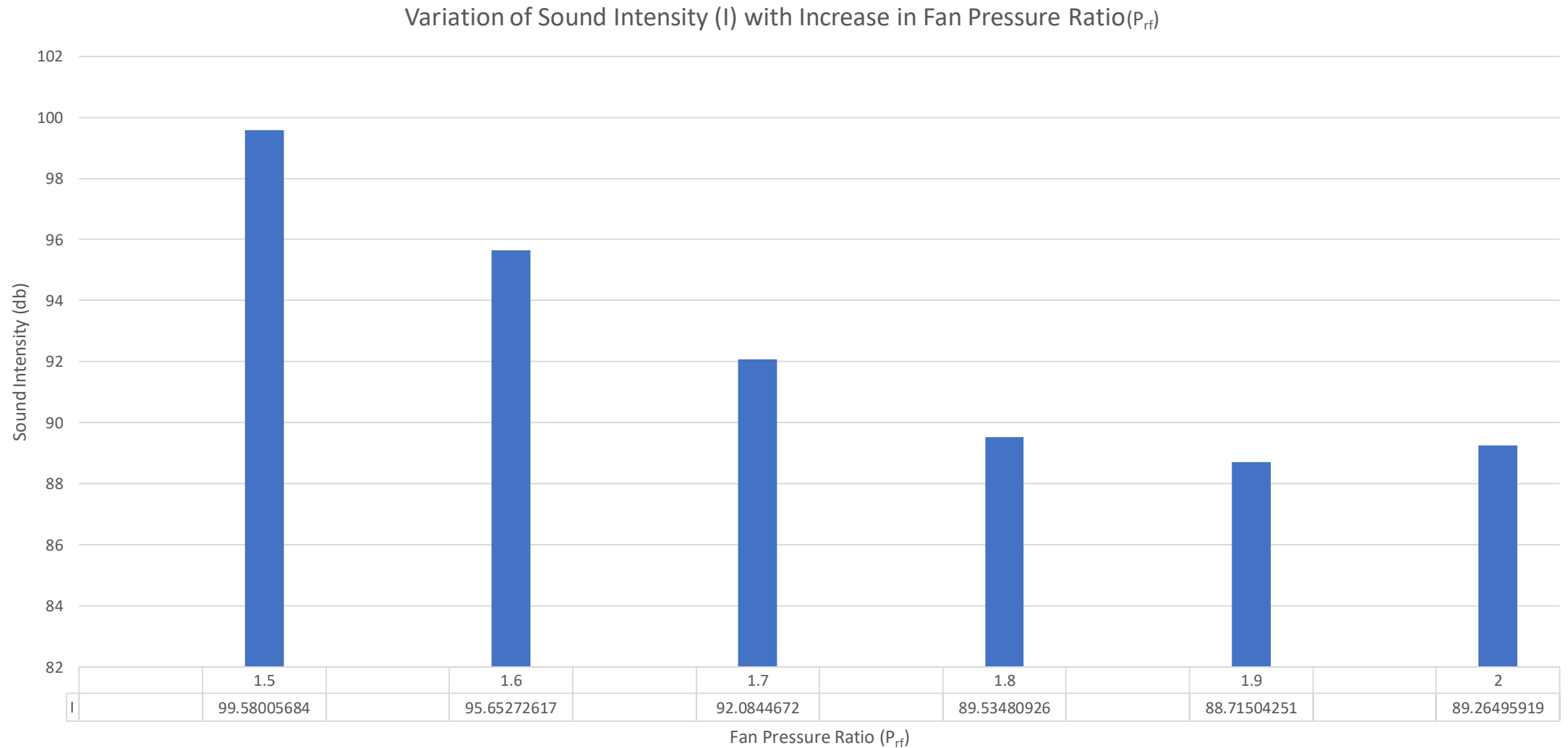
## Discussion – Sound Intensity Comparison

| Flight Condition | EPNdB<br>TFE731-3A/Lear 50 |
|------------------|----------------------------|
| Takeoff (Lab)    | 84.0 [2]                   |

Analytical Method using multiple PrF values.

| PrF | I (dB)  |
|-----|---------|
| 1.5 | 99.5801 |
| 1.6 | 95.6527 |
| 1.7 | 92.0845 |
| 1.8 | 89.5348 |
| 1.9 | 88.7150 |
| 2   | 89.2650 |

# Results – Effects of Fan Pressure Ratio on I





## Discussion – Noise Intensity

- The difference in the noise intensity between the analytical and practical method could be because of 1 or more of the reasons listed below.
  1. Ideal Gas Behavior assumption.
  2. Fan Pressure ratio assumption and optimization.
  3. Efficiency values used are from textbook reference.
  4. Noise due to other factors like aerodynamic interactions, vibrations etc. are not accounted.
  5. Fan and Core noise interaction is complicated and not addressed in the calculations.

# Further Study

1. Landing Noise calculations.
2. Consider the stage noise.

# Appendix

## 1. Class Projects

- i. Turbofan Sizing and Acoustics Analysis.

## 2. Tools used.

- Programming – python 3
- Report – Microsoft PowerPoint and Adobe reader.

# Reference List

- [1] P. G. Hill and C. R. Peterson, *Mechanics and thermodynamics of propulsion*, 2nd ed. Reading, Mass: Addison-Wesley, 1992. Table 5.1(Turbojet) & Table 5.2 (Bypass fan)
- [2] Garrett Turbine Engine Company, Ed., *Noise characteristics of the Garrett TFE731-3A turbofan engine as installed in a Gates Learjet model 50 series aircraft*. [Phoenix, Arizona: Garrett Turbine Engine Company, 1981. Ablah Library Call Number : TL671.65 .N537 1981
- [3] N. A. Cumpsty, Jet propulsion: a simple guide to the aerodynamic and thermodynamic design and performance of jet engines. in Cambridge engine technology series, no. 2. Cambridge ; New York, NY, USA: Cambridge University Press, 1997.
- [4] B. Gunston and Jane's Information Group, Eds., Jane's aero-engines. Coulsdon, Surrey, U.K. ; Alexandria, Va: Jane's Information Group, 1996.( TL701 .J36 1996  
Ablah Library)
- [5] "TFE731 Turbofan Engine.". [Honeywell's Website](#)

# Reference List – Contd..

- [6] S. Institution, “Garrett (Allied Signal) TFE731-3R Turbofan Engine,” Smithsonian Institution. [Smithsonian's Website](#)
- [7] G. A. Bobula and R. A. Lottig, *Inlet Reynolds number and temperature effects on the steady-state performance of a TFE731-2 turbofan engine*. in NASA technical memorandum, no. NASA TM X-3537. Washington : Springfield, Va: National Aeronautics and Space Administration ; for sale by the National Technical Information Service, 1977.
- [8] “Reduction of Noise at Source.” : [The International Civil Aviation Organization \(ICAO\)](#)
- [9] “Stage 5 Airplane Noise Standards,” A rule by the Federal Aviation Administration. [Federal Register - National Archives](#)
- [10]Wichita State University , Wichita and Kansas 67260978-3456, “Dr. Roy Myose Ph.D [Wichita State University Faculty](#). (AE-742 & AE-743)

# Extra Slides

## Sound Intensity Calculation Parameters [1],[2]

| Parameter | Value | Description                         | Parameter | Description                                      | Value           | Unit     |
|-----------|-------|-------------------------------------|-----------|--------------------------------------------------|-----------------|----------|
| EtaC      | 0.85  | Compressor efficiency               | PrF       | Fan pressure ratio                               | 1.5(Varied)     | -        |
| EtaF      | 0.85  | Fan efficiency                      | PrB       | Burner pressure ratio (assumed no pressure loss) | 1               | -        |
| EtaT      | 0.9   | Turbine efficiency                  | R         | Gas constant                                     | 287             | J/(kg·K) |
| EtaN      | 0.98  | Core nozzle efficiency              | QR        | Heat release                                     | 4.50E+07        | J/kg     |
| EtaNF     | 0.97  | Fan nozzle efficiency               | a0        | Speed of sound                                   | 340.2686        | m/s      |
| EtaD      | 0.97  | Diffuser efficiency                 | B         | Bypass ratio                                     | 2.8             | -        |
| gammaD    | 1.4   | Specific heat ratio for diffuser    | PrC       | Compressor pressure ratio                        | 14.6            | -        |
| gammaC    | 1.37  | Specific heat ratio for compressor  | Ma        | Mach number (assumed 0 for this calculation)     | 0               | -        |
| gammaF    | 1.4   | Specific heat ratio for fan         | T04       | Turbine inlet temperature                        | 1133.15(Varied) | K        |
| gammaB    | 1.35  | Specific heat ratio for burner      | Pa        | Ambient pressure                                 | 96647.53        | Pa       |
| gammaT    | 1.33  | Specific heat ratio for turbine     | Ta        | Ambient temperature                              | 288.706         | K        |
| gammaN    | 1.36  | Specific heat ratio for core nozzle |           |                                                  |                 |          |
| gammaNF   | 1.4   | Specific heat ratio for fan nozzle  |           |                                                  |                 |          |

Table 3 : TFE731 Noise Calculation Parameters

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# Sound Intensity Calculation – Results(Lowest I)

| Parameter    | Description                                 | Value    | Unit           |
|--------------|---------------------------------------------|----------|----------------|
| B            | Bypass ratio                                | 2.8      | -              |
| PrC          | Compressor pressure ratio                   | 14.6     | -              |
| T04          | Turbine inlet temperature                   | 1220.49  | K              |
| f            | Fuel-air ratio                              | 0.014476 | -              |
| Core, V      | Core nozzle exit velocity                   | 341.2845 | m/s            |
| Fan, V       | Fan nozzle exit velocity                    | 341.4527 | m/s            |
| Spec. Thrust | Specific thrust (Thrust Per Unit Mass Flow) | 1302.293 | N/(kg/s)       |
| Thrust Prod  | Total thrust production                     | 16458.41 | N              |
| TSFC         | Thrust-specific fuel consumption            | 1.11E-05 | kg/(N·s)       |
| Acore        | Core area                                   | 0.106041 | m <sup>2</sup> |
| SICore       | Core sound intensity level                  | 82.90458 | dB             |
| SIfan        | Fan sound intensity level                   | 87.39329 | dB             |
| I            | Total sound intensity level                 | 88.71504 | dB             |
| P05          | Stagnation pressure at turbine outlet       | 129806.9 | Pa             |
| T05          | Stagnation temperature at turbine outlet    | 729.7124 | K              |